

DC-DC Converter Electrical & Efficiency Test Report

Galvanic Isolation Hipot, Step Load Dynamics, and UVLO Ramp

1. Step Load Transient Response & Hipot Isolation

- Test Case TC-DCDC-021: Transient step load test: 10 A to 200 A transient caused 13.84 V output, recovered within 320 μ s. Verdict: P
- Test Case TC-DCDC-022: Hipot test between HV and LV domains applied 3000 V AC rms for 60 s; measured leakage current was 0.3

2. Efficiency Mapping Across Load & Voltage Ripple

- Test Case TC-DCDC-023: Efficiency measured at 1000 W (95.2%) and 2000 W (94.8%). Low power (300 W) and full power (3000 W)
- Test Case TC-DCDC-024: Output ripple measured at 25°C ambient was 68.0 mVpp. Thermal chamber ripple measurements at -40°C

3. Undervoltage Lockout (UVLO) Shutdown Trip Point

- Test Case TC-DCDC-026: Input voltage ramp test: DC-DC converter continued switching down to 290.0 V DC before tripping, failing th